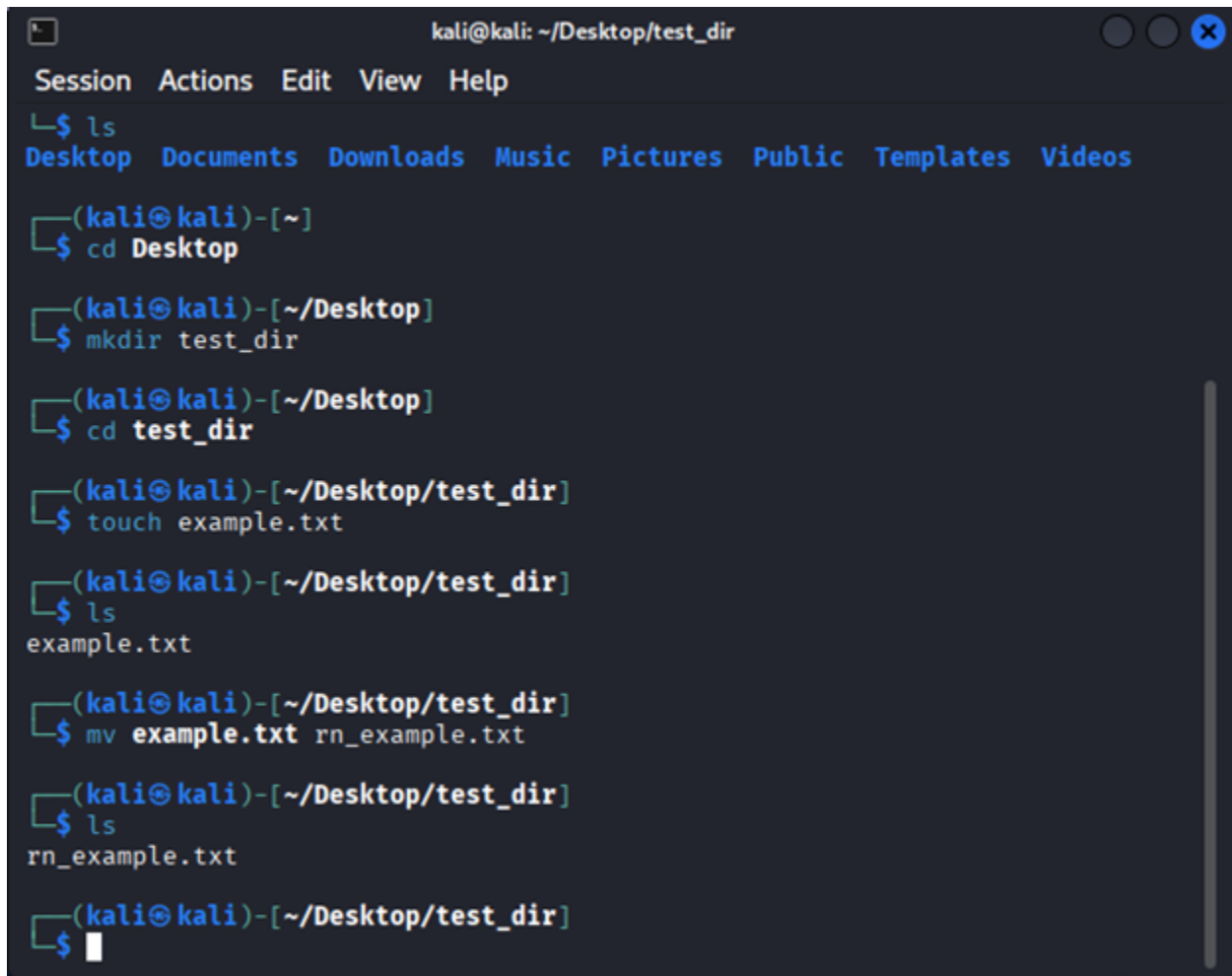


Creating and Renaming Files/Directories

Create a directory named test_dir using mkdir.

Inside test_dir, create an empty file called example.txt.

Rename example.txt to renamed_example.txt using mv



```
kali@kali: ~/Desktop/test_dir
Session Actions Edit View Help
└─$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos

(kali@kali)-[~]
└─$ cd Desktop

(kali@kali)-[~/Desktop]
└─$ mkdir test_dir

(kali@kali)-[~/Desktop]
└─$ cd test_dir

(kali@kali)-[~/Desktop/test_dir]
└─$ touch example.txt

(kali@kali)-[~/Desktop/test_dir]
└─$ ls
example.txt

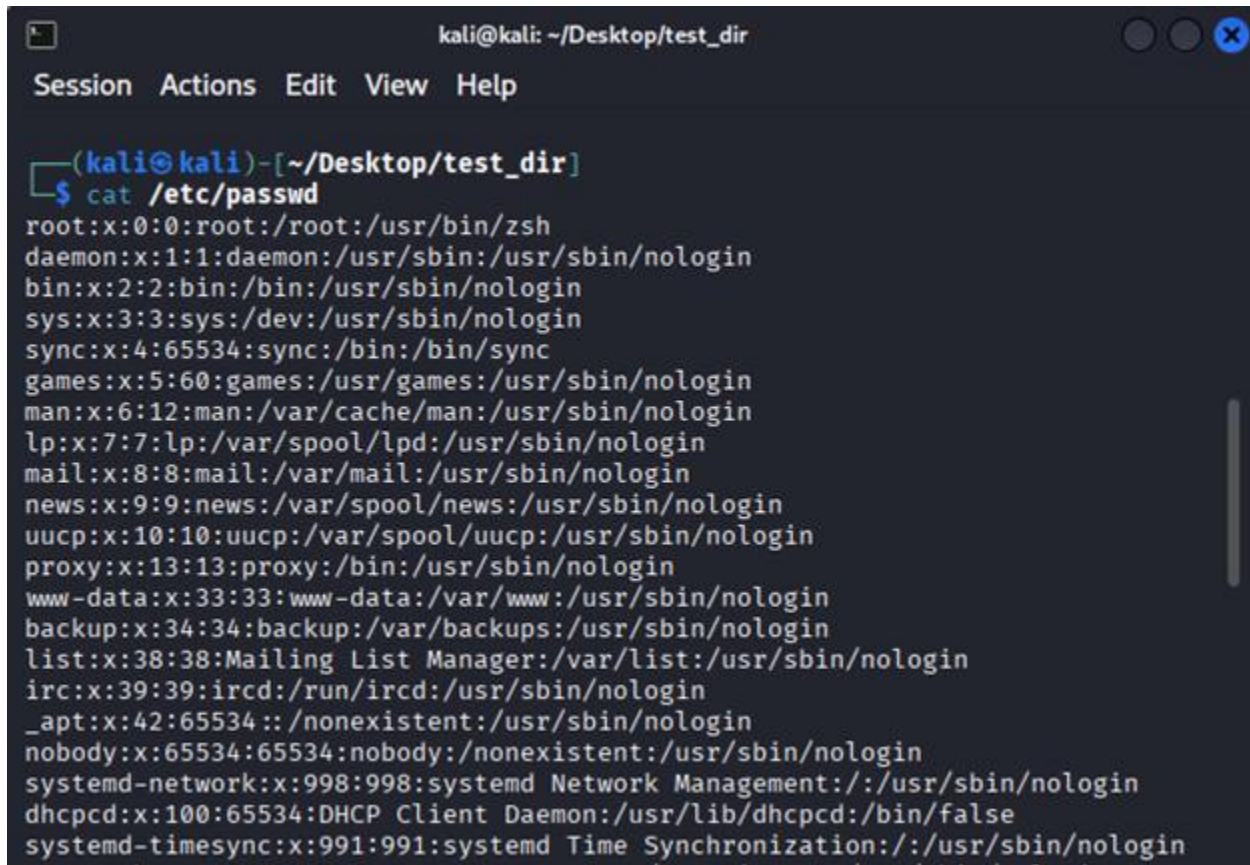
(kali@kali)-[~/Desktop/test_dir]
└─$ mv example.txt rn_example.txt

(kali@kali)-[~/Desktop/test_dir]
└─$ ls
rn_example.txt

(kali@kali)-[~/Desktop/test_dir]
└─$
```

2. Viewing File Contents

Use cat to display the contents of /etc/passwd.

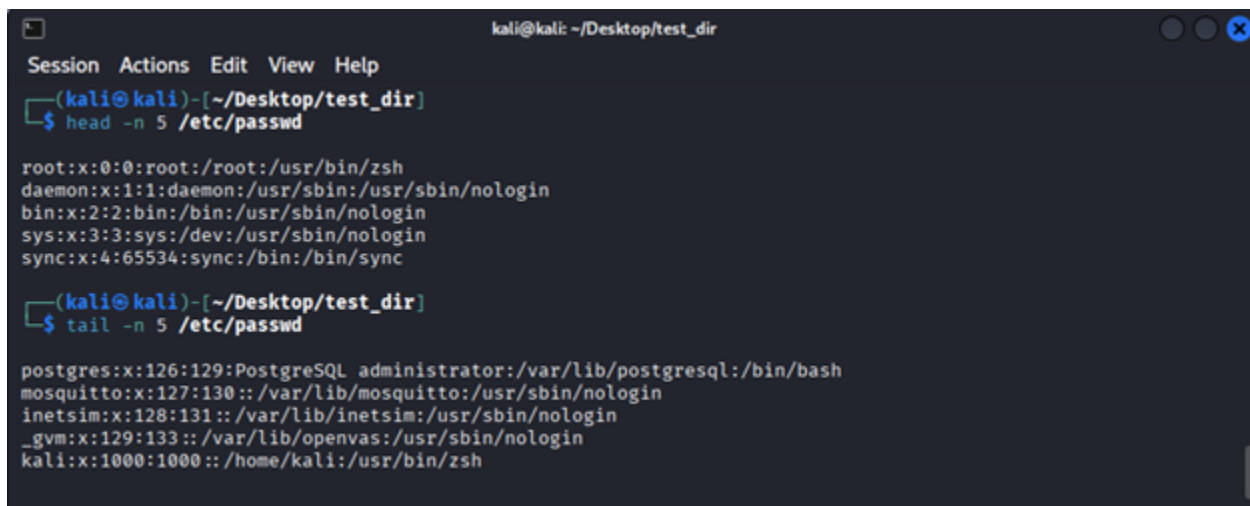


```
kali@kali: ~/Desktop/test_dir
Session Actions Edit View Help

(kali@kali)-[~/Desktop/test_dir]
$ cat /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
systemd-timesync:x:991:991:systemd Time Synchronization:/:/usr/sbin/nologin
```

Display only the first 5 lines of /etc/passwd using head.

Display only the last 5 lines of /etc/passwd using tail.



```
kali@kali: ~/Desktop/test_dir
Session Actions Edit View Help

(kali@kali)-[~/Desktop/test_dir]
$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync

(kali@kali)-[~/Desktop/test_dir]
$ tail -n 5 /etc/passwd
postgres:x:126:129:PostgreSQL administrator:/var/lib/postgresql:/bin/bash
mosquitto:x:127:130::/var/lib/mosquitto:/usr/sbin/nologin
inetsim:x:128:131::/var/lib/inetsim:/usr/sbin/nologin
_gvm:x:129:133::/var/lib/openvas:/usr/sbin/nologin
kali:x:1000:1000::/home/kali:/usr/bin/zsh
```

3. Searching for Patterns

Use grep to find all lines containing the word "root" in /etc/passwd.

```
(kali@kali)-[~/Desktop/test_dir]
$ grep "Tushar" /etc/passwd

Tushar:x:1001:1001:tushar p,12,1234567890,0987654321:/home/Tushar:/bin/bash
```

4. Zipping and Unzipping

Compress the test_dir directory into a file named test_dir.zip using zip.

Unzip test_dir.zip into a new directory named unzipped_dir.

```
kali@kali: ~/Desktop

Session Actions Edit View Help

(kali@kali)-[~/Desktop]
$ zip -r test_dir.zip test_dir

adding: test_dir/ (stored 0%)
adding: test_dir/rn_example.txt (stored 0%)

(kali@kali)-[~/Desktop]
$ ls
test_dir  test_dir.zip

(kali@kali)-[~/Desktop]
$ unzip test_dir.zip -d unzipped_dir

Archive: test_dir.zip
  creating: unzipped_dir/test_dir/
  extracting: unzipped_dir/test_dir/rn_example.txt

(kali@kali)-[~/Desktop]
$ ls
test_dir  test_dir.zip  unzipped_dir
```

5. Downloading Files

Use wget to download a file from a URL (e.g., <https://example.com/sample.txt>).

```
(kali@kali)-[~/Desktop]
$ wget https://example.com/sample.txt
--2025-12-10 23:17:39--  https://example.com/sample.txt
Resolving example.com (example.com)... 23.192.228.80, 23.192.228.84, 23.215.0.136, ...
Connecting to example.com (example.com)|23.192.228.80|:443 ... connected.
HTTP request sent, awaiting response... 404 Not Found
2025-12-10 23:17:40 ERROR 404: Not Found.
```

6. Changing Permissions

Create a file named `secure.txt` and change its permissions to read-only for everyone using `chmod`.

```
(kali㉿kali)-[~/Desktop]
$ touch secure.txt

(kali㉿kali)-[~/Desktop]
$ chmod 444 secure.txt

(kali㉿kali)-[~/Desktop]
$ ls
secure.txt  test_dir  test_dir.zip  unzipped_dir
```

7. Working with Environment Variables

Use `export` to set a new environment variable called `MY_VAR` with the value "Hello, Linux!".

```
(kali㉿kali)-[~/Desktop]
$ export MY_VAR="Hello, Linux!"
dquote>
dquote> echo $MY_VAR
dquote>

(kali㉿kali)-[~/Desktop]
$ echo $MY_VAR
```